Name: MD. Rokib Hasan Oli

ID: 2211950642

Practice (Lab 4)

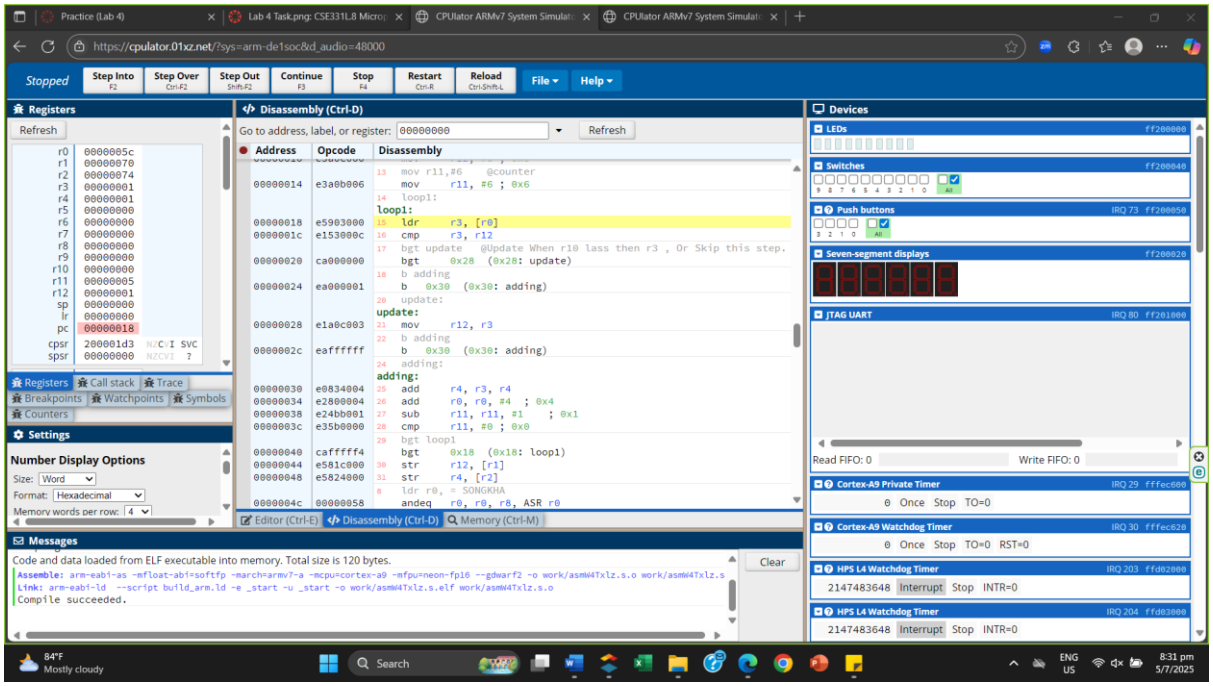
Q1

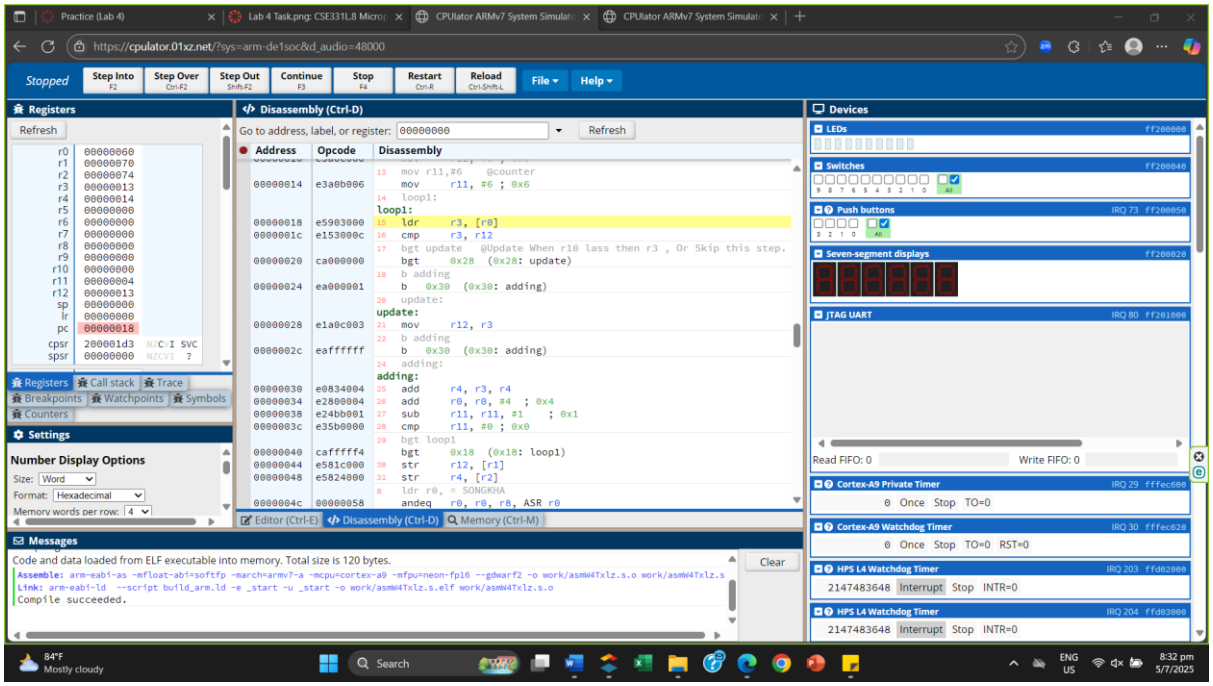
```
.global _start
.data
    SONGKHA: .word 1,0x13,5,4,0x12,0x17
    array_min: .word 0
    array_sum: .word 0
.text
_start:
   ldr r0, = SONGKHA
   ldr r1, = array_min
   ldr r2, = array_sum
          mov r4,#0    @Sum --> value 0
   mov r12,#0   @MAX --> Value 0
   mov r11,#6    @counter
   loop1:
     ldr r3,[r0]
     cmp r3,r12
     bgt update   @Update When r10 lass then r3 , Or Skip this step.
     b adding

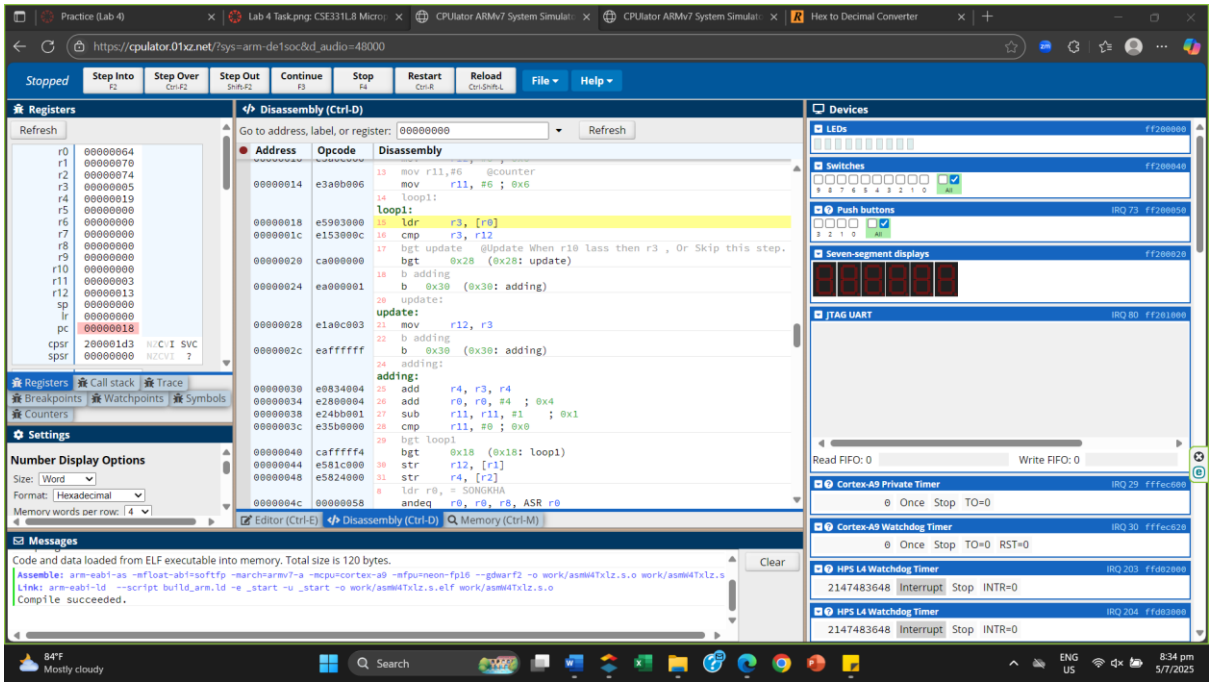
          update:
                mov r12,r3
                b adding

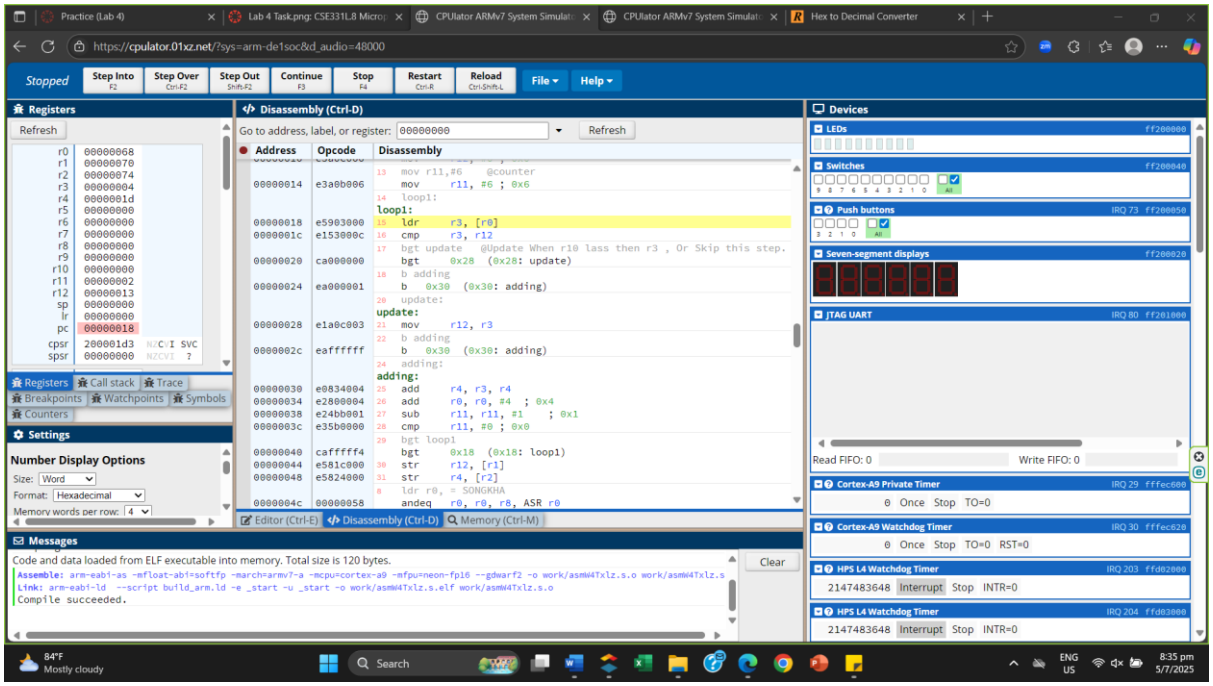
          adding:
```

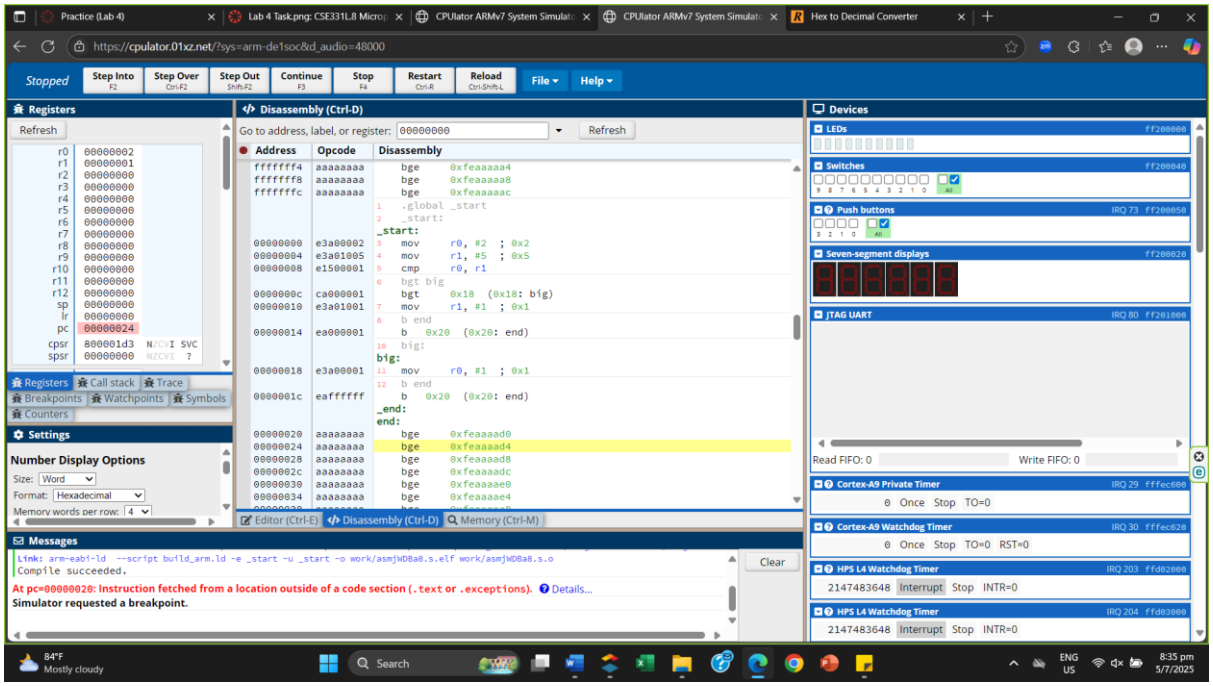
```
        add r4,r3,r4
        add r0,r0,#4
        sub r11,r11,#1
        cmp r11,#0
bgt loop1
    str r12,[r1]
    str r4,[r2]
.end
```

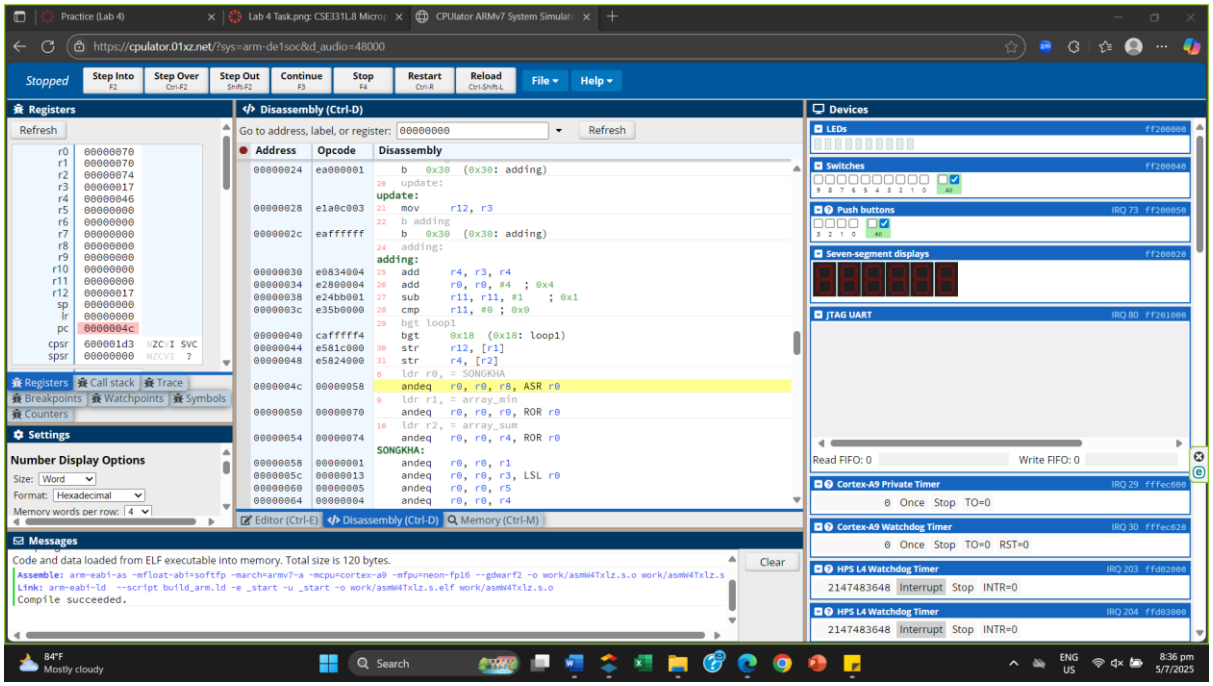




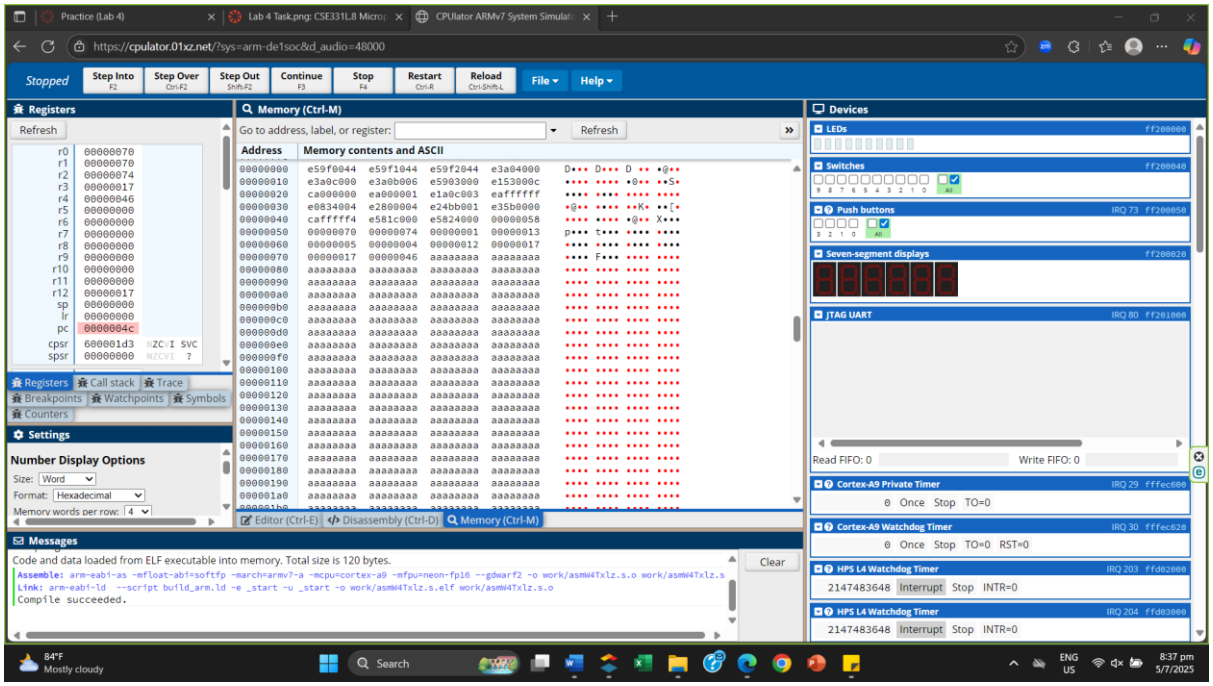

Practice (Lab 4)
Lab 4 Task.png: CSE331L8 Micro
CPUlator ARMv7 System Simulat
CPUlator ARMv7 System Simulat
Hex to Decimal Converter
https://cpulator.01xz.net/?sys=arm-de1soc&d_audio=48000
Stopped
Step Into F2
Step Over Ctrl-F2
Step Out Shift-F2
Continue F3
Stop F4
Restart Ctrl-R
Reload Ctrl-Shift-L
File
Help
Registers
Refresh
r0 00000064
r1 00000070
r2 00000074
r3 00000005
r4 00000019
r5 00000000
r6 00000000
r7 00000000
r8 00000000
r9 00000000
r10 00000000
r11 00000003
r12 00000013
sp 00000000
lr 00000000
pc 00000018
cpsr 200001d3 N:0 Z:0 C:1 I SVC
spsr 00000000 N:0 Z:0 V:1 ?
Registers
Call stack
Trace
Breakpoints
Watchpoints
Symbols
Counters
Settings
Number Display Options
Size: Word
Format: Hexadecimal
Memory words per row: 4
Disassembly (Ctrl-D)
Go to address, label, or register: 00000000
Refresh
Address
Opcode
Disassembly
13 mov r11,#6 @counter
00000014 e3a0b006 mov r11, #6 ; 0x6
14 loop1:
loop1:
00000018 e5903000 15 ldr r3, [r0]
0000001c e153000c 16 cmp r3, r12
17 bgt update @Update When r10 lass then r3 , Or Skip this step.
00000020 ca000000 bgt 0x28 (0x28: update)
18 b adding
00000024 ea000001 b 0x30 (0x30: adding)
20 update:
update:
00000028 e1a0c003 21 mov r12, r3
22 b adding
0000002c eaffffff b 0x30 (0x30: adding)
24 adding:
adding:
00000030 e0834004 25 add r4, r3, r4
00000034 e2800004 26 add r0, r0, #4 ; 0x4
00000038 e24bb001 27 sub r11, r11, #1 ; 0x1
0000003c e35b0000 28 cmp r11, #0 ; 0x0
29 bgt loop1
00000040 cafffff4 bgt 0x18 (0x18: loop1)
00000044 e581c000 30 str r12, [r1]
00000048 e5824000 31 str r4, [r2]
8 ldr r0, = SONGKHA
0000004c 00000058 andeq r0, r0, r8, ASR r0
Editor (Ctrl-E)
Disassembly (Ctrl-D)
Memory (Ctrl-M)
Messages
Code and data loaded from ELF executable into memory. Total size is 120 bytes.
Assemble: arm-eabi-as -mfloat-abi=softfp -march=armv7-a -mcpu=cortex-a9 -mfpu=neon-fp16 --gdwarf2 -o work/asmW4Txlz.s.o work/asmW4Txlz.s
Link: arm-eabi-ld --script build_arm.ld -e _start -u _start -o work/asmW4Txlz.s.elf work/asmW4Txlz.s.o
Compile succeeded.
Clear
Devices
LEDs ff200000
Switches ff200040
Push buttons IRQ 73 ff200050
Seven-segment displays ff200020
JTAG UART IRQ 80 ff201000
Read FIFO: 0
Write FIFO: 0
Cortex-A9 Private Timer IRQ 29 fffec600
0 Once Stop TO=0
Cortex-A9 Watchdog Timer IRQ 30 fffec620
0 Once Stop TO=0 RST=0
HPS L4 Watchdog Timer IRQ 203 ffd02000
2147483648 Interrupt Stop INTR=0
HPS L4 Watchdog Timer IRQ 204 ffd03000
2147483648 Interrupt Stop INTR=0
84°F
Mostly cloudy
Search
ENG
US
8:34 pm
5/7/2025


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Step Over Ctrl-F2
Step Out Shift-F2
Continue F3
Stop F4
Restart Ctrl-R
Reload Ctrl-Shift-L
File
Help
Registers
Refresh
r0 00000068
r1 00000070
r2 00000074
r3 00000004
r4 0000001d
r5 00000000
r6 00000000
r7 00000000
r8 00000000
r9 00000000
r10 00000000
r11 00000002
r12 00000013
sp 00000000
lr 00000000
pc 00000018
cpsr 200001d3 N:0 Z:0 C:1 I SVC
spsr 00000000 N:0 Z:0 V:1 ?
Registers
Call stack
Trace
Breakpoints
Watchpoints
Symbols
Counters
Settings
Number Display Options
Size: Word
Format: Hexadecimal
Memory words per row: 4
Disassembly (Ctrl-D)
Go to address, label, or register: 00000000
Refresh
Address
Opcode
Disassembly
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00000014 e3a0b006 mov r11, #6 ; 0x6
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loop1:
00000018 e5903000 15 ldr r3, [r0]
0000001c e153000c 16 cmp r3, r12
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00000020 ca000000 bgt 0x28 (0x28: update)
18 b adding
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20 update:
update:
00000028 e1a0c003 21 mov r12, r3
22 b adding
0000002c eaffffff b 0x30 (0x30: adding)
24 adding:
adding:
00000030 e0834004 25 add r4, r3, r4
00000034 e2800004 26 add r0, r0, #4 ; 0x4
00000038 e24bb001 27 sub r11, r11, #1 ; 0x1
0000003c e35b0000 28 cmp r11, #0 ; 0x0
29 bgt loop1
00000040 cafffff4 bgt 0x18 (0x18: loop1)
00000044 e581c000 30 str r12, [r1]
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8 ldr r0, = SONGKHA
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Link: arm-eabi-ld --script build_arm.ld -e _start -u _start -o work/asmW4Txlz.s.elf work/asmW4Txlz.s.o
Compile succeeded.
Clear
Devices
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Switches ff200040
Push buttons IRQ 73 ff200050
Seven-segment displays ff200020
JTAG UART IRQ 80 ff201000
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Write FIFO: 0
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0 Once Stop TO=0
Cortex-A9 Watchdog Timer IRQ 30 fffec620
0 Once Stop TO=0 RST=0
HPS L4 Watchdog Timer IRQ 203 ffd02000
2147483648 Interrupt Stop INTR=0
HPS L4 Watchdog Timer IRQ 204 ffd03000
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84°F
Mostly cloudy
Search
ENG
US
8:35 pm
5/7/2025


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Step Into F2
Step Over Ctrl-F2
Step Out Shift-F2
Continue F3
Stop F4
Restart Ctrl-R
Reload Ctrl-Shift-L
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Registers
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r0 00000002
r1 00000001
r2 00000000
r3 00000000
r4 00000000
r5 00000000
r6 00000000
r7 00000000
r8 00000000
r9 00000000
r10 00000000
r11 00000000
r12 00000000
sp 00000000
lr 00000000
pc 00000024
cpsr 800001d3 N ZCV I SVC
spsr 00000000 NZCVI ?
Registers
Call stack
Trace
Breakpoints
Watchpoints
Symbols
Counters
Settings
Number Display Options
Size: Word
Format: Hexadecimal
Memory words per row: 4
Disassembly (Ctrl-D)
Go to address, label, or register: 00000000
Refresh
Address
Opcode
Disassembly
fffffff4 aaaaaaaa bge 0xfeaaaaa4
fffffff8 aaaaaaaa bge 0xfeaaaaa8
fffffffc aaaaaaaa bge 0xfeaaaaac
1 .global _start
2 _start:
00000000 e3a00002 3 mov r0, #2 ; 0x2
00000004 e3a01005 4 mov r1, #5 ; 0x5
00000008 e1500001 5 cmp r0, r1
6 bgt big
0000000c ca000001 bgt 0x18 (0x18: big)
00000010 e3a01001 7 mov r1, #1 ; 0x1
8 b end
00000014 ea000001 b 0x20 (0x20: end)
10 big:
big:
00000018 e3a00001 11 mov r0, #1 ; 0x1
12 b end
0000001c eafffffff b 0x20 (0x20: end)
_end:
end:
00000020 aaaaaaaa bge 0xfeaaaad0
00000024 aaaaaaaa bge 0xfeaaaad4
00000028 aaaaaaaa bge 0xfeaaaad8
0000002c aaaaaaaa bge 0xfeaaaadc
00000030 aaaaaaaa bge 0xfeaaaae0
00000034 aaaaaaaa bge 0xfeaaaae4
Editor (Ctrl-E)
Disassembly (Ctrl-D)
Memory (Ctrl-M)
Devices
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2147483648 Interrupt Stop INTR=0
HPS L4 Watchdog Timer IRQ 204 ffd03000
2147483648 Interrupt Stop INTR=0
Messages
Clear
Link: arm-eabi-ld --script build_arm.ld -e _start -u _start -o work/asmjWD8a8.s.elf work/asmjWD8a8.s.o
Compile succeeded.
At pc=00000028: Instruction fetched from a location outside of a code section (.text or .exceptions). Details...
Simulator requested a breakpoint.
84°F
Mostly cloudy
Search
ENG
US
8:35 pm
5/7/2025


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Step Over Ctrl-F2
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Continue F3
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Registers
Refresh
r0 00000070
r1 00000070
r2 00000074
r3 00000017
r4 00000046
r5 00000000
r6 00000000
r7 00000000
r8 00000000
r9 00000000
r10 00000000
r11 00000000
r12 00000017
sp 00000000
lr 00000000
pc 0000004c
cpsr 600001d3 nZCv I SVC
spsr 00000000 NZCVI ?
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22 b adding
0000002c eafffffff b 0x30 (0x30: adding)
24 adding:
adding:
00000030 e0834004 25 add r4, r3, r4
00000034 e2800004 26 add r0, r0, #4 ; 0x4
00000038 e24bb001 27 sub r11, r11, #1 ; 0x1
0000003c e35b0000 28 cmp r11, #0 ; 0x0
29 bgt loop1
00000040 cafffff4 bgt 0x18 (0x18: loop1)
00000044 e581c000 30 str r12, [r1]
00000048 e5824000 31 str r4, [r2]
8 ldr r0, = SONGKHA
0000004c 00000058 andeq r0, r0, r8, ASR r0
9 ldr r1, = array_min
00000050 00000070 andeq r0, r0, r0, ROR r0
10 ldr r2, = array_sum
00000054 00000074 andeq r0, r0, r4, ROR r0
SONGKHA:
00000058 00000001 andeq r0, r0, r1
0000005c 00000013 andeq r0, r0, r3, LSL r0
00000060 00000005 andeq r0, r0, r5
00000064 00000004 andeq r0, r0, r4
Editor (Ctrl-E)
Disassembly (Ctrl-D)
Memory (Ctrl-M)
Devices
LEDs ff200000
Switches ff200040
Push buttons IRQ 73 ff200050
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Link: arm-eabi-ld --script build_arm.ld -e _start -u _start -o work/asmW4Txlz.s.elf work/asmW4Txlz.s.o
Compile succeeded.
84°F
Mostly cloudy
Search
ENG
US
8:36 pm
5/7/2025

Memory



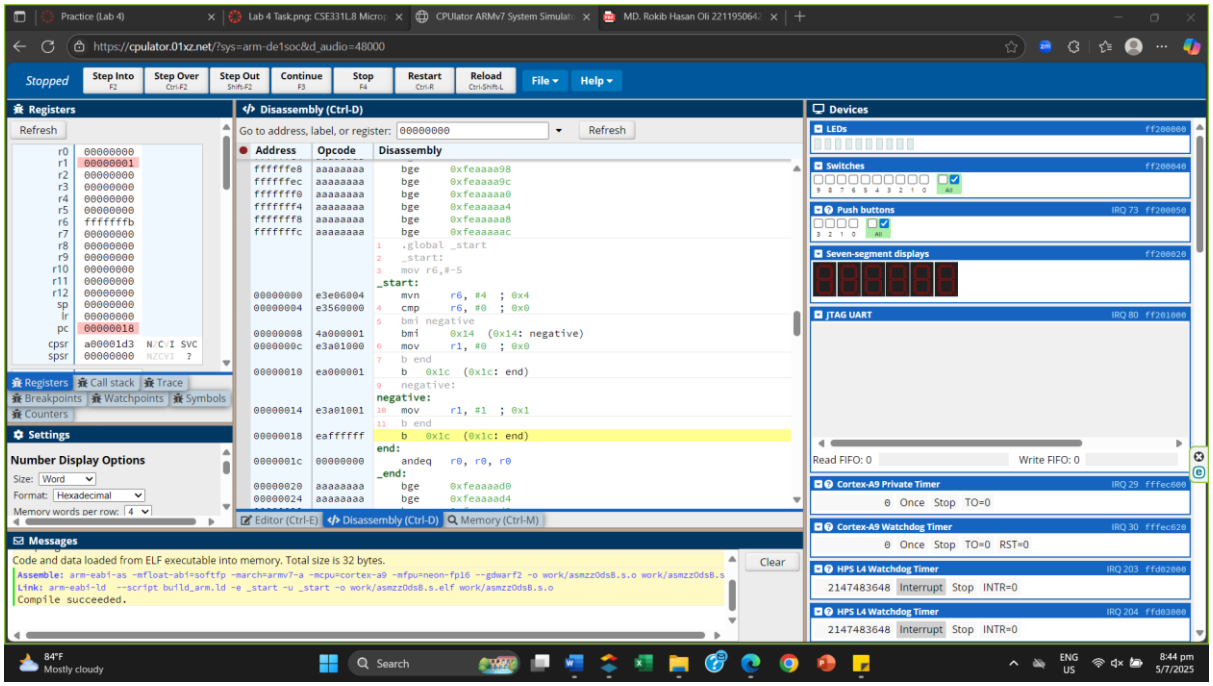
Q2

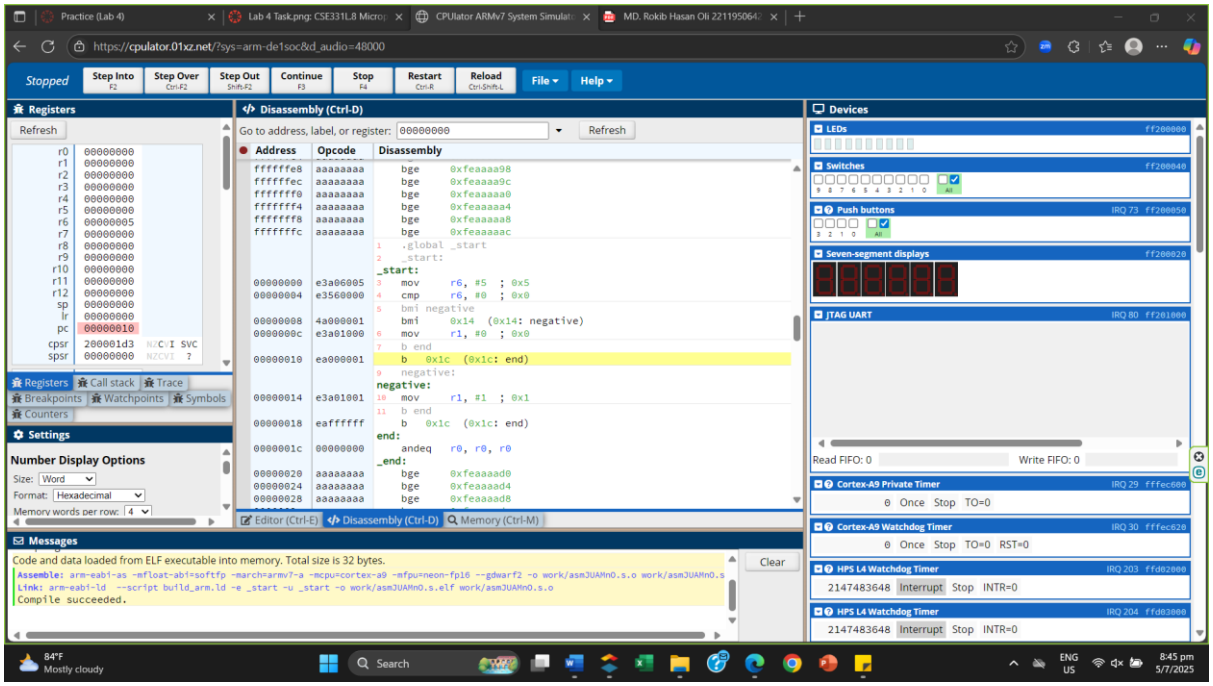
```
.global _start

_start:
        mov r6,#-5
        cmp r6,#0
        bmi negative
        mov r1,#0
        b end

negative:
        mov r1,#1
        b end

end:
        .end
```





Q3

CPU later don't support function.

```
.global _start

_start:

        mov r1,#3

        bl 8_diya_vaag
```

```
8_diya_vaag:
lsr r1,r1,#3
cmp r1,#0
beq check

mov r8,#3
b end

check:
mov r8,#2
b end

end:
.end
```

